# Chapter 10
# Coastal Modeling

In this chapter we discuss how to simulate inundation and flooding in coastal landscapes, explain workflows for developing storm surge and sea level rise scenarios, and present two case studies. The first case study is a series of storm surge and dune breach scenarios for a populated barrier island. The second is a design—informed by flood and erosion modeling—for resilient, sustainable guest housing at a coastal research institute.

## 10.1 Modeling Potential Inundation

Inundation and flooding in coastal landscapes is caused by storm surge and sea level rise. It is very complex and depends upon topography and bathymetry, tides, sea level surge and waves, and wind and rainfall. The spatial extent of flooding can be approximated by "spreading" the observed or predicted water level throughout the landscape (Clinch et al. 2012).

To approximate inundation and flooding we can spread the given water level from a predefined seed located in the ocean using the module *r.lake*. This approach respects the obstacles, so if a foredune protects a low lying area, this area will not be flooded even if the water level exceeds its elevation. We describe the algorithm as follows: an increase in water level approximates the dynamics of inundation, but we assume that the water spreads at an infinite speed and thus floods everything that it can reach instantly. Therefore this method is best used to map the maximum possible extent assuming that there are no changes in topography during the process.

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A. Petrasova et al., *Tangible Modeling with Open Source GIS*,
DOI 10.1007/978-3-319-25775-4_10

## 10.2 Case Study: Simulating Flood Risks for Barrier Island Communities

We explored the impact of topography on inundation and flooding at the Jockey's Ridge sand dune complex in Nags Head on the Outer Banks barrier islands of North Carolina (Fig. 10.1) with a series of a storm surge scenarios. We carved breaches into the foredune of our sand model to identify which homes were most vulnerable to storms.

For this study we selected a study area including the beach, foredune, homes, road, dune complex, maritime forest, and soundside shore. We derived a bare earth DEM and a DSM with buildings and vegetation for our study from a 2009 lidar survey. Then we cast a polymeric sand model of the DEM from a CNC routed mold. We projected an orthophotograph, elevation data, the shoreline, and building footprints as context. Based on the building footprints we added massing models of the buildings to our sand model.

To study the potential impacts of a hurricane we simulated storm surge at increasing levels from 1 to 4 m using the add-on module *r.lake.series*. Then we breached the foredune at a point where it has been compromised by the construction of boardwalk for beach access and compared the extent of flooding at different surge levels (Fig. 10.2).

```
r.lake.series elevation=elev_2009_1m output=flood \
    start_water_level=1 end_water_level=4 water_level_step=0.1 \
    coordinates=913525,249507
g.gui.animation strds=flood
```

**Fig. 10.1** The Jockey's Ridge dune complex in Nags Head, North Carolina

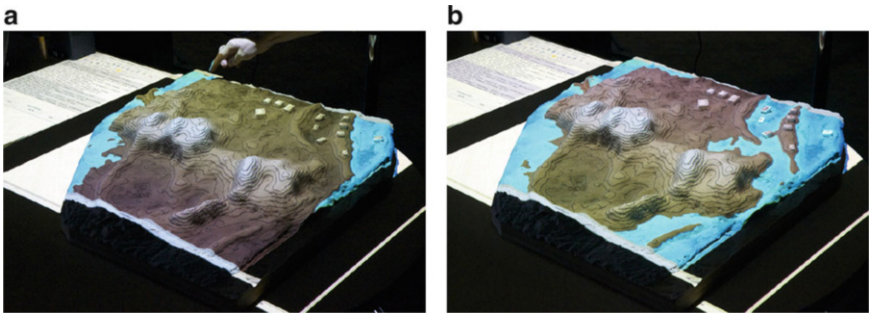


**Fig. 10.2** Inundation extent simulations for Jockey's Ridge area for (**a**) 1 m and (**b**) 2 m of storm surge

## 10.3 Case Study: Designing Resilient Coastal Architecture

Two architecture students—David Koontz and Faustine Pastor—used Tangible Landscape to site a series of residential pavilions (Fig. 10.3) for visitors staying at the UNC Coastal Studies Institute on Roanoke Island, North Carolina as part of an joint architecture-engineering studio. They used storm surge and water flow simulations to inform the selection of building sites. Then they spaced pilings for the buildings and adaptively graded a biofiltration swale based on near real-time feedback about water flow, sediment flux, and erosion-deposition.[1]

The UNC Coastal Studies Institute—the site for the studio—is a research institute on Roanoke Island in the Outer Banks of North Carolina. Students in this North Carolina State University studio course led by Professors Joe DeCarolis, David Hill, and Ranji Ranjithan were tasked with designing sustainable housing for guests and researchers visiting the institute.[2] The site is at risk of inundation from sea level rise and storm surge and is in a sensitive back-barrier salt marsh.

Given the risk of flooding David and Faustine used Tangible Landscape to visualize different levels of inundation and test the vulnerability of potential building sites. They placed basswood massing models of buildings on a cast sand model of the landscape and then used the module *r.lake* to simulate 4 m storm surge and check if their buildings would stay dry (Fig. 10.4). In their design the buildings would be raised on piers and connected by boardwalks over a bioswale. They simulated water flow to find the right spacing for the piers over swale. Then they adaptively graded the swale using feedback from water flow, erosion-deposition, and sediment flux simulations, sculpting the channel so that water and sediment flowed towards a bioretention pond (Fig. 10.5).

[1]Watch the video on Youtube: https://youtu.be/PbbzWymGvLo

[2]Read more about the studio on the Coastal Dynamics Design Lab website: http://design.ncsu.edu/coastal-dynamics-design-lab/

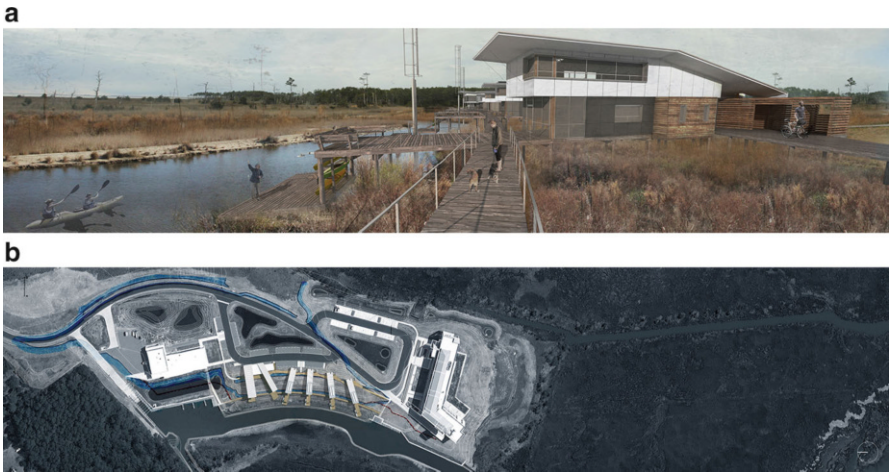


**Fig. 10.3** The students' design for a series of pavilions for the UNC Coastal Studies Institute on Roanoke Island, NC. Images by David Koontz and Faustine Pastor. (**a**) A rendering of the proposed design for a residential pavilion. (**b**) A masterplan of the UNC Coastal Studies Institute with the proposed residential pavilions

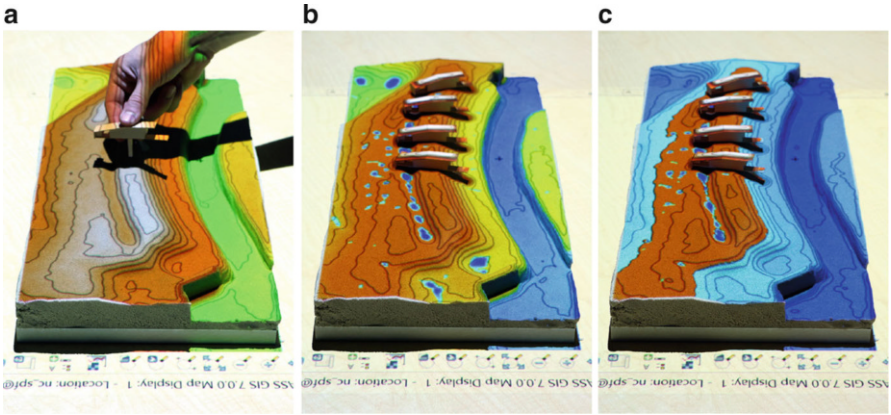


**Fig. 10.4** Students using water flow and flood simulations to adaptively inform the placement of a series of pavilions for the Coastal Studies Institute on Roanoke Island, NC: (**a**) siting buildings by hand on an artificial ridge, (**b**) studying the flow of water around the buildings and (**c**) testing whether the buildings stay dry with 4 m storm surge

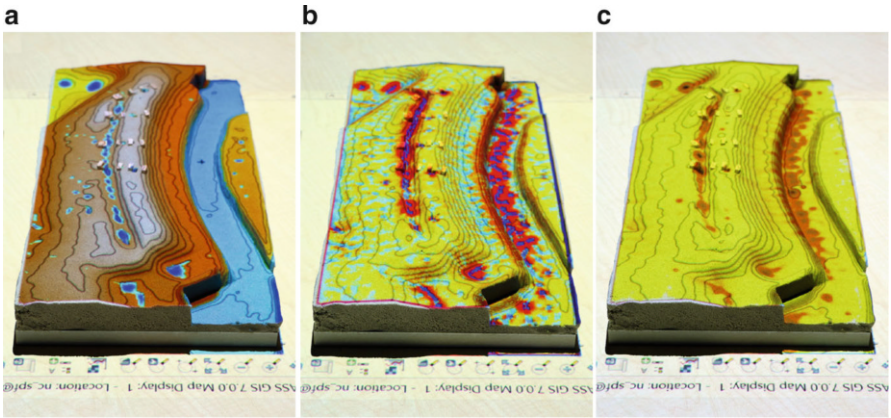


**Fig. 10.5** Students using (**a**) water flow, (**b**) erosion-deposition, and (**c**) sediment flux simulations to space structural piers and adaptively grade a biofiltration swale for the Coastal Studies Institute on Roanoke Island, NC

## Reference

Clinch, A. S., Russ, E. R., Oliver, R. C., Mitasova, H., and Overton, M. F. (2012). Hurricane Irene and the Pea Island breach: prestorm site characterization and storm surge estimation using geospatial technologies. *Shore & Beach*, 80(2):38–46.